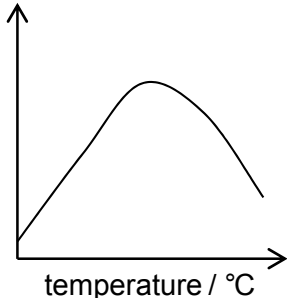
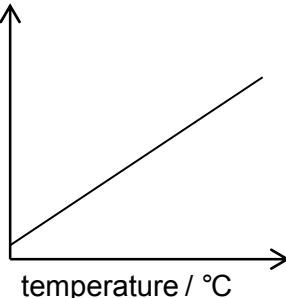


1(a)(i)	as temperature increases, the carbon dioxide increases / increases and then decreases/levels off / AW ;	1
1(a)(ii)	<i>independent variable:</i> temperature ; <i>dependent variable:</i> colour / pH ;	2
1(a)(iii)	any 7 of: 1 <i>ref. to</i> setting up at least 5 water-baths at different temperatures ; 2 <i>ref. to</i> a suitable range of temperatures with units ; 3 same / stated / known, volume / concentration of hydrogencarbonate / indicator (solution) to each test-tube ; 4 <i>ref. to</i> adding oxygen into indicator solution ; 5 <i>ref. to</i> indicator starting at same colour or pH in all test-tubes ; 6 <i>ref to</i> foil, mix, (immediately) start timing / mix, (immediately) foil, (immediately) start timing ; 7 use same / stated / known, volume of <i>Chlorella</i> ; 8 <i>ref. to</i> putting indicator and <i>Chlorella</i> in separate tubes in a water-bath to reach desired temperature / fixed time ; 9 mix <i>Chlorella</i> and indicator and <i>ref. to</i> a specific end point to be measured ; 10 (<i>ref. to</i> a control using), killed / boiled / dead, <i>Chlorella</i> (of same volume) ; 11 (if time / pH) at least three replicates / repeats and finding mean (of time or pH) or identify / eliminate / remove anomalies or (if colour) <i>ref. to</i> using at least three replicates / repeats ; 12 <i>ref. to</i> low risk ;	7

Question	Answer	Marks
1(b)(i)	<u>1</u> time (to reach standard colour / pH) ;	1
1(b)(ii)	temperature on x-axis ; labelled axis and units ; appropriate curve for named axis ; e.g. rate of respiration (s⁻¹ or min⁻¹)  temperature / °C rate of respiration (s⁻¹ or min⁻¹)  temperature / °C	3
1(c)(i)	0.31 ;	1
1(c)(ii)	there is no, (significant) correlation / relationship, between the cell density and the rate of carbon dioxide production or the correlation / relationship between cell density and the rate of carbon dioxide production is not significant ;	1
1(c)(iii)	strong ; positive correlation ;	2

Question	Answer	Marks
1(c)(iv)	<i>any 2 of:</i> 1 correlation does not mean causation ; 2 fewer than number of paired observations (recommended) for Pearson test ; 3 no replicates / no repeats / only done experiment once ; 4 only narrow range of cell densities ; 5 bacteria / other microorganisms present that are respiring ;	2